NHS
North Central London
Integrated Care Board

Headache Primary Care Clinical Pathway (Adult)

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NCL-Wide / Borough(s)	NCL-Wide
Author(s)	See note

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Key

- Pi "Must do" actions for GP's
- B Recommendations for Primary Care
- R Red flag / urgent referral
- O Routine referral
- P Public health intervention
- G Audio-visual aids for patients and GP
- Click icon for clinical evidence
- Right-click to use hyperlink

Note: Alpha-numeric step references are to aid printing in black & white and colour blindness

Investigations must be within 6 months of date of referral

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With acknowledgment to the NHNN Headache Group

1.0 (B) Patient presents with headache

2.0 (B) Assessment
History (see boxes below)
Blood Pressure
Neurological Examination (including fundoscopy)
If uncertain during assessment, seek Neurology advice via Consultant Connect

2.1 (G)

3.0 (R) Red flags

No → Yes

No

4.0 (B) Migraine

5.0 (B) Diagnosis
2 or more of:
- Unilateral headache although often bilateral
- Lasting 4-72 hours
- Pulsating/ throbbing
- Nausea and/or vomiting
- Light and sound sensitivity
- +/- aura

6.0 (B) Tension-type headache

7.0 (B) Diagnosis
- Bilateral location, pressing or tightening pain, mild or moderate intensity
- Not aggravated by routine physical activities

If any migrainous features present, then manage as migraine.

8.0 (B) Cluster headache

9.0 (B) Diagnosis
- Very severe unilateral orbital, supraorbital or temporal pain
- Autonomic features ipsilateral to pain e.g. lacrimation, conjunctival injection, eyelid swelling/droop or nasal congestion
- Associated agitation during episodes e.g. pacing behaviour

10.0 (B) Trigeminal/Facial Neuralgia

11.0 (B) Diagnosis
- Severe, lancinating, electric shock pain
- Triggered by touch, washing face, eating or cold air +/- autonomic features
- Lasting a few seconds to 2 minutes
- 1-50 attacks per day

12.0 (B) Diagnosis not clear

13.0 (G) Seek Neurology advice and guidance via ERS/Consultant Connect/ discussion with on-call neurologist

Consider and address lifestyle factors** 20% settle by reducing triggers

22.0 (B)

23.0 (Pi) Offer combination therapy [5], [7] and in early headache phase of attack (not aura phase)

1st line: Triptan + Simple analgesic + Prokinetic
Antiemetic (do not prescribe opioids e.g. Co-codamol).

Each combination should be trialled for 3 headache days. If no response in 2 of 3 trials then change the triptan and try again for at least 3 headaches. Two different triptans should each be trialled on a minimum of two separate occasions, before determining whether patient is a triptan non-responder.

2nd line: Rimegepant (if 2 triptans ineffective or not tolerated or contraindicated) Acute treatments should not be taken on ≥ 10 days/month to avoid medication overuse headache. [5]

Headache diary [2]

23.1 (B)

24.0 (Pi) Management
- If medication is needed, amitriptyline* or nortriptyline may be trialled (see dose regimen in migraine pathway)
- Consider acupuncture for chronic pain

Headache diary [2]

24.1 (B)

25.0 (Pi) Management
Acute:
- Sumatriptan* 6mg s/c (max 12mg/ day). Can be used daily for the duration of the cluster bout.
- 2nd line: Sumatriptan* or zolmitriptan nasal spray ipsilaterally (off-label in cluster headache - for patients who cannot tolerate sumatriptan injection)

Continue oxygen therapy following **specialist initiation**.

Prevention:
Continue verapamil 80mg TDS following **specialist initiation**, increase every 2 weeks in steps of 80mg to 240mg TDS. ECG before each dose increase.

25.0 (B)

26.0 (Pi) Management
- Carbamazepine 100mg up to twice daily, titrate in steps of 100-200mg every 2 weeks maximum 1600mg daily
- FBC, U&E, LFT, TFT monitoring is advised
- Modified release preparations may be useful at night if the person experiences breakthrough pain.
- Once pain is in remission, the dosage should be gradually reduced to the lowest possible maintenance level.

Yes

14.0 (R) Sudden onset (<5min to max severity) **or** Systemically unwell – fever, meningism, rash, BP >180/110 **or** Focal neurological signs or altered mental state **or** New headache with recent head trauma **or** Pregnant or recently pregnant **or** Cancer or immunocompromised

15.0 (R) Urgent same day clinical review

16.0 (R) Symptoms and signs suggestive of giant cell arteritis [1]

17.0 (R) Same day discussion with on-call Rheumatologist/ medical registrar

18.0 (R) Symptoms and signs of acute narrow angle glaucoma (headache and red eye)

19.0 (R) Same day discussion with on-call Ophthalmologist

20.0 (R) If progressive subacute loss of central neurological function **or** Progressive, subacute loss of central neurological function **or** Headache with features suggestive of raised intracranial pressure e.g. worse on supine position, triggered by valsalva **or** New onset daily persistent headache duration <12 weeks

21.0 (R) Same day discussion with on-call Neurologist/medical team for consideration of urgent neurology appointment or urgent 2ww referral to the brain and CNS cancer pathway if the patient meets the criteria [2]

Recommended medicines for TREATMENT of acute migraine

Type	Name	Recommended dosage
Simple analgesics	Naproxen	250-500mg OD-BD (max 1000mg daily)
	Ibuprofen	200-400mg TDS (max 1200mg daily)
	Aspirin	600-900mg (max 3600mg daily)
	Paracetamol	1g QDS (max 4g daily)
Antiemetics	Prochlorperazine	5-10mg (max 10mg every 8hrs)
	Metoclopramide	10mg (max 10 mg every 8hrs) (in patients over 20yr old)
	Domperidone	10mg (max 10mg every 8 hrs)
Triptans – first line	Sumatriptan	Oral: 50-100mg stat (max 300mg daily)
Triptans - second line	Zolmitriptan	2.5 and 5mg oral tablets or melts (max 10mg daily) 5mg nasal spray (max 10mg per day)
	Rizatriptan	Oral: 10mg (max 20mg daily; 10mg daily if on propranolol)
	Naratriptan	1mg and 2.5mg tablets (max 5mg per day)
	Sumatriptan	Intranasal: 10-20mg stat (max 40mg daily)
CGRP receptor antagonist	Rimegepant	75mg daily (for treatment of migraine) 75mg on alternate days (for prevention of migraine)

27.0 (G) Seek advice & guidance via ERS/Consultant Connect if queries

Refer to Neurology OPD. Speak to on-call Neurologist if any concerns

28.0 (O)

29.0 (B)

Consider a preventive treatment for patients who:
- Experience migraine symptoms on 4 or more days per month (for more than 3 months)
- Less than 4 migraine days per month, but with poor response to acute treatment or cannot take suitable acute treatment for migraine attacks due to contraindications or intolerance

Side-effects are common with preventive treatments, but often improve over time; titrate medication as tolerated/required [7] (page 26-27 for further guidance) to target dose for a minimum of 2 months before determining effectiveness (≥50% in episodic and ≥30% in chronic). Review treatment every six months. Once patient satisfied with control consider tapering the dose after 6-12 months and advise patient to monitor for deterioration.

This approach will also mitigate the risk of escalation of acute treatment and consequent development of medication overuse headache***. Acute treatment on more than 2 days per week is associated with medication overuse, which renders preventive treatment less effective.

29.1

30.0 (Pi)

Migraine continued
Preventative medication
Aim: To reduce frequency and severity of migraine
1st Line
- Propranolol 40mg nocte for 2 weeks. Increase to 40mg twice daily for two weeks and 80mg twice daily thereafter. Maximum 80-160mg/day or
- Metoprolol 25mg twice daily and increase by 25mg up to maximum dose of 100mg twice daily.

2nd Line
- Amitriptyline 10mg at night for 2 - 4 weeks increasing by 10mg every 2 -4 weeks according to response and tolerability. Maximum 1mg/kg/day or 75mg once daily if weight >75mg. Consider nortriptyline *off label* if amitriptyline has been effective but patient has experienced intolerable side effects of drowsiness. Recommended doses for nortriptyline the same as for amitriptyline.
- Topiramate 25mg nocte and increase every two weeks to a target dose of 50mg twice daily. Further increase can be made following discussion with a neurologist (**Patient must be registered with pregnancy prevention programme (PPP) and associated requirements completed**).

3rd Line
- Candesartan (off label) starting dose 4 mg twice daily, increase weekly by 4 mg up to 8 mg twice daily; review at 3 months

NOTE: There is insufficient evidence for the safety of these drugs in pregnancy or breastfeeding. See *Drug Considerations box below for safety and monitoring information

- If patient has migraine with aura, stop combined oral contraceptive pill. Progesterone only pill can be offered to patients with migraine with aura
- Address lifestyle factors**
- Preventative treatment should be tried for at least 3 months at the maximum tolerated dose, before deciding whether or not it is effective. A good response to treatment is defined as a 50% reduction in the severity and frequency of migraine attacks. A review of ongoing preventative medication should be considered after 6–12 months; treatment can be gradually withdrawn in many patients.

31.0 (O)

Consider referral to secondary care (General Neurology) if patient has failed to respond to optimised doses/has not tolerated two or more migraine therapies NB Advanced migraine therapies (e.g. BOTOX , CGRP inhibitors for preventative treatment) will be considered for patients who meet specified NICE criteria including failure or intolerance of 3 or more preventative drug treatments.
Consider referral to a specialist headache service for advice and consideration of further treatment when patients have exhausted at least 3 treatment options above due to lack of efficacy or tolerability issues

31.0 (B)

***Drug Considerations**

Prescribing decisions should be made with reference to the patient's current clinical situation and their future plans (e.g. pregnancy or contraception)
- **Triptans**: Avoid concomitant use with SSRIs. Avoid in people who have had a previous MI, or have ischaemic heart disease, stroke, peripheral vascular disease, uncontrolled hypertension.
- **Amitriptyline**: Avoid in people who have had recent MI or who have arrhythmia. Use with caution in ischaemic heart disease or epilepsy. Common adverse effects include constipation, weight gain, dry mouth, sedation, nausea, and dizziness. These are usually apparent in the first couple of weeks of taking the drug and settle with continued use.
- **Beta-blockers**: Avoid beta-blockers in asthma, COPD, diabetes (if associated with frequent episodes of hypoglycaemia), severe peripheral arterial disease, or uncontrolled heart failure.
- **Candesartan**: Associated with foetal abnormalities, avoid in pregnancy. Ensure the patient is using highly effective contraception. Monitor renal function and serum potassium.
- **Topiramate**: Associated with significant harm to the unborn child. Harms included a higher risk of congenital malformation, low birth weight and a potential increased risk of intellectual disability, autistic spectrum disorder and attention deficit hyperactivity disorder in children of mothers taking topiramate during pregnancy. Topiramate is now contraindicated in pregnancy and in women of childbearing potential unless the conditions of a Pregnancy Prevention Programme are fulfilled. MHRA Topiramate new safety measures June 2024. There is an increased risk of metabolic acidosis with topiramate in patients with conditions that predispose to acidosis e.g. hepatic or renal impairment or diarrhoea.
- **Carbamazepine**: Can be prescribed generically for headache indications. Associated with an increased risk of major congenital malformations hence should not be used in women of childbearing potential unless the benefit is judged to outweigh the risks. Can impair the effectiveness of hormonal contraceptives; women of childbearing potential should use highly effective contraception during treatment with carbamazepine and for at least two weeks after stopping treatment. Carbamazepine is a potent enzyme inducer, potential interactions with the patient's other medication should be considered. Serious cutaneous reactions including toxic epidermal necrolysis (TEN) and Stevens-Johnson syndrome (SJS) have been reported during treatment with carbamazepine – risk higher in certain populations (e.g. individuals of Han Chinese or Thai origin). Counsel patients on how to recognise how to recognise signs of blood, liver, or skin disorders, and advise to seek immediate medical attention if symptoms such as fever, rash, mouth ulcers, bruising, or bleeding develop.
- **Rimegepant:** for supporting information see Appendix 1

For more information regarding safety of AEDs in pregnancy, please consult MHRA Drug Safety Update: Antiepileptic drugs in pregnancy: updated advice following comprehensive safety review [4].
The above information on contraindications and cautions for medication use in headache is not exhaustive. For further details of contraindications, special warnings and precautions for use of the medicines included in the pathway, please refer to the manufacturer's Summary of Product Characteristics (SmPC) https://www.medicines.org.uk/emc/ and current BNF https://bnf.nice.org.uk/

****Lifestyle factors:**
20% settle by reducing triggers (eg stress, dehydration, missed meals, poor sleep). Regular exercise may be beneficial. Evaluate and address predisposing factors eg stress, depression or anxiety, menstruation, menopause, head or neck trauma.

***** Consider medication overuse if use of acute analgesics is > 10 days a month**
Management: Stop analgesia completely. Offer antiemetics and offer NSAIDS up to twice a week (or no analgesia at all)
Note: pain killer medication taken regularly for non-headache pain, such as joint or back pain, can result in medication overuse headaches. There is no difference in outcome with either abrupt or gradual withdrawal of the causative drug.

See EBICS policy for other alternative treatment

Patient information
OUCH UK and oral information about headache disorders, NHS patient information on cluster headaches. e.g. BASH Patient Leaflet - Common Headache Problems The Migraine Trust (for people living with migraine); OUCH UK (support for people living with cluster headaches); NHS patient information on cluster headaches.
Explain the risk of medication overuse headache to people who are using acute treatments for their headache disorder.
Consider referral to IAPT if stress is a significant factor.

This pathway has been developed from published guidance: NICE CG 150 NICE CG150 Headaches in over 12s: diagnosis and management [5], and NICE NG12 Suspected cancer: recognition and referral [6] in collaboration with local neurologists. This guidance is to assist GPs with clinical decision making and is not intended to replace clinical judgment.

[1]: NICE CKS - Giant cell arteritis. Last revised in March 2022. Available via https://cks.nice.org.uk/topics/giant-cell-arteritis/
[2]: The Migraine Trust. Keeping a Headache Diary. Available via https://migrainetrust.org/live-with-migraine/self-management/keeping-a-migraine-diary/#page-section-2
[3]: MHRA Drug Safety Update. Topiramate (Topamax): start of safety review triggered by a study reporting an increased risk of neurodevelopmental disabilities in children with prenatal exposure. Published 21 July 2022. Available via https://www.gov.uk/drug-safety-update/topiramate-topamax-start-of-safety-review-triggered-by-a-study-reporting-an-increased-risk-of-neurodevelopmental-disabilities-in-children-with-prenatal-exposure
[4]: MHRA Drug Safety Update. Antiepileptic drugs in pregnancy: updated advice following comprehensive safety review. Published 7 January 2021. Available via https://www.gov.uk/drug-safety-update/antiepileptic-drugs-in-pregnancy-updated-advice-following-comprehensive-safety-review
[5]: NICE CG150Headaches in over 12s: diagnosis and management. Published: 19 September 2012 Last updated: 17 December 2021, Available: https://www.nice.org.uk/guidance/cg150
[6]: NICE guideline [NG12]. Suspected cancer: recognition and referral. Published: 23 June 2015 Last updated: 15 December 2021. Available via https://www.nice.org.uk/guidance/ng12/chapter/Recommendations-organised-by-site-of-cancer#brain-and-central-nervous-system-cancers
[7]: British Association for the Study of Headache. (BASH). National Headache Management System for Adults 2019. Available via https://headache.org.uk/wp-content/uploads/2023/02/bash-guideline-2019.pdf
[8]: NICE TA919. Rimegepant for treating migraine. Published: 18 October 2023 Available: https://www.nice.org.uk/guidance/ta919
[9]: NICE TA906. Rimegepant for preventing migraine. Published: 5 July 2023. Available https://www.nice.org.uk/guidance/ta906

Appendix 1 – Rimegepant oral lyophilisate (Vydura® ▼) prescribing information

Rimegepant selectively binds with high affinity to the human calcitonin generelated peptide (CGRP) receptor and antagonises CGRP receptor function. Elevated blood concentrations of CGRP have been associated with migraine attacks.

Rimegepant is licensed for

- Acute treatment of migraine with or without aura in adults;
- Preventive treatment of episodic migraine in adults who have at least 4 migraine attacks per month.

Two NICE technology appraisals have been published in relation to the licensed indications for rimegepant.

NICE technology appraisal (TA) 919 recommends rimegepant as an option for the acute treatment of migraine with or without aura in adults, only if for previous migraines:

- at least 2 triptans were tried and they did not work well enough or
- triptans were contraindicated or not tolerated, and nonsteroidal anti-inflammatory drugs (NSAIDs) and paracetamol were tried but did not work well enough.

NICE technology appraisal (TA) 906 recommends rimegepant as an option for preventing episodic migraine in adults only if:

- They have at least 4 and fewer than 15 migraine attacks per month AND
- At least 3 preventative treatments have not worked. Treatment should be stopped after 12 weeks if the frequency of migraine attacks does not reduce by at least 50%

NICE advise that rimegepant could be initiated by a specialist then later be prescribed by a GP in primary care

Prescribing information – refer to the SmPC for full prescribing information

Dose and administration

Acute treatment of migraine
The recommended dose is 75 mg rimegepant, as needed, once daily.
Prophylaxis of migraine
The recommended dose is 75 mg rimegepant every other day.

- The maximum dose per day is 75 mg rimegepant.
- Another dose of rimegepant should be avoided within 48 hours when it is concomitantly administered with moderate inhibitors of CYP3A4 or with strong inhibitors of P-gp (e.g. cyclosporine, verapamil, quinidine)
- There is limited experience with rimegepant in patients aged 65 years or older.
- No dose adjustment is required in older patients, those with renal impairment or mild to moderate hepatic impairment. Rimegepant should be avoided in those with severe hepatic impairment
- The oral lyophilisate should be placed on the tongue or under the tongue. It will disintegrate in the mouth and can be taken without liquid.
- Patients should be advised to use dry hands when opening the blister and referred to the package leaflet for complete instruction

Contraindications

- Hypersensitivity to the active substance or to any of the excipients

Warning and Precautions

Rimegepant is not recommended:

- in patients with severe hepatic impairment
- in patients with end-stage renal disease (CLcr < 15 ml/min)
- for concomitant use with strong inhibitors of CYP3A4 (e.g., clarithromycin, itraconazole, ritonavir)
- for concomitant use with strong (e.g., phenobarbital, rifampicin, St John's wort) or moderate (e.g., bosentan, efavirenz, modafinil) inducers of CYP3A4

Overuse of any type of medicinal products for headaches can make them worse. If this situation is experienced or suspected, treatment should be discontinued.

Assessment/review of treatment

No special monitoring is required in terms of safety profile.

When prescribed for prevention:
The patient should keep a headache diary to record the number of migraine days per month.

After 12 weeks, it should be assessed for efficacy by the initiating specialist, Patients can continue treatment in primary care if this assessment determines that the frequency of migraine attacks has reduced by at least 50%.

When to stop Rimegepant:

Prevention: Stop treatment after 12 weeks, if the frequency of migraine attacks does not reduce by at least 50%

Acute: Consider stopping treatment if migraine has not responded to a trial of 2 doses for 2 separate attacks

Adverse Effects

The most common adverse reaction was nausea for acute treatment (1.2%) and for migraine prophylaxis (1.4%). Most of the reactions were mild or moderate in severity. Hypersensitivity, including dyspnoea and severe rash, occurred in less than 1% of patients treated

References

Pfizer UK Ltd. Summary of Product Characteristics for Vydura 75mg oral lyophilisate. Last updated on emc 13 Jul 23. Accessed 19 Feb 24